

AI Drawing-to-ERP Compliance Gateway

Read, verify and push construction drawings into RealSoft - automatically

Project Overview | Coral Business Solutions / RealSoft ERP | Prepared by the ERP RealSoft team

1. What is ERP RealSoft

ERP RealSoft is an AI gateway that turns approved construction drawings into clean, validated ERP records. A user uploads a drawing (PDF or image); ERP RealSoft reads it with AI, extracts the title-block and floor-plan details, checks them against a rule book, and sends the verified data straight into RealSoft - with a full report and **no manual typing**.

Everything happens live on screen: as the drawing is processed the user watches each step stream in real time, then sees a pass / warning / fail verdict, the extracted fields with confidence scores, and a report with the original drawing **marked up in red where mistakes are found**.

2. The problems we solve

Problem today	How ERP RealSoft solves it
Manual data entry from drawings into the ERP is slow and error-prone	AI reads the title block and floor plan and fills the ERP fields automatically.
Drawings reach the ERP incomplete or unapproved	An 18-rule engine checks required fields, approval stamp, formats and confidence before any push.
Hard to trust AI output blindly	Every field carries a confidence score; low-confidence and failed items are flagged for human review.
Reviewers cannot see WHERE a drawing is wrong	The report draws red boxes on the drawing at the exact problem locations.

Unreadable / blurry scans waste time and AI cost	A blur gate rejects too-blurry uploads up front with a clear message - no wasted processing.
No audit trail or shareable record	Every drawing, exception, correction and ERP push is stored; HTML / PDF / Excel reports are generated.

3. How it works (the pipeline)

From upload to ERP push, each step streams live to the screen and must complete well under 60 seconds:

Step	What happens
1. Upload + checks	File received and stored; pre-upload rules verify format, size and readability.
2. Quality gate	Image sharpness is measured; too-blurry drawings are stopped with a note.
3. AI extraction	Claude Vision reads the title block, floor plan, dimensions, materials and stamps.
4. Validation	18 rules (R01-R18) check completeness, confidence and formats; a verdict is produced.
5. ERP mapping	Extracted data is normalised into the RealSoft DrawingMaster format.
6. ERP push	The record is sent to RealSoft (or simulated until credentials are provided).
7. Report	A branded report is generated with the red-marked drawing, fields and validation results.

4. AI integrations in the project

ERP RealSoft never calls a model directly from the app - all inference is routed through a pluggable AI provider that prefers a hosted gateway, falls back to a local sidecar, and finally to built-in demo data, so the product always runs.

AI capability	What it does
Claude Vision extraction	Anthropic Claude reads the drawing image and returns the structured fields + confidence scores + bounding boxes for each field.
AI Gateway (AI-as-a-Service)	A hosted sidecar (Hostinger VPS) runs the model server-side and exposes tasks: DRAWING_EXTRACT, DRAWING_ANALYZE, DRAWING_VALIDATE, ERP_MAP, REPORT.

Deterministic prepass	For PDFs with a text layer (or via OCR), fields are read for free before the AI call and treated as high-confidence ground truth.
Confidence calibration	Raw AI confidence is re-scored using prepass agreement, format validity and cross-field consistency, so the scores can be trusted.
Blur detection	Computer vision (Laplacian-variance sharpness) decides if an image is too blurry to extract reliably.
AI red-marking	The model returns each field's location; the report draws red boxes on failed / warning fields at those spots.
ERP value mapping	The gateway can also normalise extracted fields into the exact RealSoft field shape (ERP_MAP task).

5. Features referenced from Pratyaya

Pratyaya is our sibling AI compliance-verification product. ERP RealSoft reuses several of its proven, production-tested patterns rather than re-inventing them:

From Pratyaya	How ERP RealSoft uses it
Dark navy + orange design system	The same visual identity (deep navy background, orange accents) across ERP RealSoft's UI and reports.
Red-lining on drawings	Coordinate-driven red boxes + labels that mark issues directly on the drawing, by severity.
AI-as-a-Service gateway pattern	Inference runs behind a hosted gateway instead of embedding model keys in the app.
Next.js + secure BFF architecture	The new web app proxies the backend through its own API layer and keeps the login token in a secure cookie (not in the browser).
Specialist / findings presentation	Clear verdict banners, confidence chips and a findings list, adapted to construction drawings.

6. The validation rule book (R01-R18)

Category	Checks
Pre-upload (R01-R03)	File type, file size, and that the file is readable / non-empty.

Required fields (R04-R11)	Drawing number, project name, revision, approval stamp, dimensions, materials, quantities, author/checker.
Confidence (R12-R15)	Key fields must meet confidence thresholds; very low confidence forces manual review.
Format / business (R16-R18)	Drawing-number, revision and quantity values must match expected formats.

A "Strict mode" can treat warnings as errors for stricter sign-off.

7. RealSoft ERP integration

ERP RealSoft sends each verified drawing into the RealSoft DrawingMaster module. Three connection methods are supported (see the separate "RealSoft Integration Options" document for details):

- **Live API link** (recommended) - real-time push over a secure web API. Code is built; needs URL + key.
- **Excel / CSV import** via RealData Hub - simple batch fallback; needs the import template.
- **Read-back check** via the Data API - optional, to confirm pushes and avoid duplicates.

Until RealSoft credentials are provided, ERP RealSoft runs in a safe **simulation mode** - it prepares the exact payload it would send, without sending anything live.

8. Architecture and technology

Layer	Technology
Web app (UI)	Next.js (React) deployed on Vercel - fast, modern, public URL. DEPLOYED
Backend API	Python FastAPI with live event streaming, on a VPS - runs the AI pipeline, validation and reporting.
AI gateway	Hosted sidecar (Hostinger VPS) running the vision model server-side.
Data	SQLite store for drawings, extractions, exceptions, corrections and ERP pushes.
Reports	Branded HTML + PDF + Excel, generated server-side.

9. Security

- Login with signed tokens (JWT) and hashed passwords (bcrypt); brute-force lockout on repeated failures.
- The web app keeps the login token in a secure, http-only cookie - it is never exposed to browser scripts or URLs.

- All backend calls go through the app's own server layer, so model and ERP secrets stay on the server.
- Inputs are validated at every boundary; the AI provider output is treated as untrusted and re-checked.

10. What we need from the client

To move ERP RealSoft from simulation to a fully live, production deployment, we need the following from Coral / RealSoft:

[**RealSoft test environment URL** - the base web address of the RealSoft API.
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[**RealSoft API key / token** - so ERP RealSoft can connect securely.
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[**Exact "import" endpoint path** - the address that accepts new drawing records.
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[**DrawingMaster field names** - the exact ERP column names so the data maps perfectly.
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[**Excel / CSV import template** - only if the Excel route (RealData Hub) is preferred.
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[**5-10 sample approved drawings** - real examples to test and tune extraction accuracy.
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[**AI provider decision** - confirm using the hosted AI Gateway, or provide an Anthropic API key.
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[**Sign-off on the validation rules and thresholds** - confirm R01-R18 match your approval policy.
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[**Production hosting details** - a domain (with HTTPS) for the backend so the live site is fully secured.
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11. Current status

- Full pipeline working end to end: upload, blur gate, AI extraction, 18-rule validation, ERP mapping, reports with red mistake markings.
- New Next.js web app **deployed live on Vercel**; the Python backend runs on the VPS.
- Running in **SIMULATION / DEMO** AI and ERP mode until RealSoft credentials and a live AI key are provided (see section 10).